

MATHEMATICS
Introduction to Modern Algebra I
Quiz 7.

1) State the First Homomorphism Theorem.

Solution: Let $\phi : G \rightarrow G'$ be a surjective homomorphism with Kernel K . Then $G' \cong G/K$ where the isomorphism is defined by $\psi(Kg) = \phi(g)$.

2) Use the First Homomorphism Theorem to prove that $\mathbb{C}^*/S^1 \cong \mathbb{R}_{>0}$ where S^1 is the multiplicative subgroup of $\mathbb{C}^*$ defined by $S^1 = \{z \in \mathbb{C}^* \text{ such that } |z| = 1\}$.

Solution: Let $\phi : \mathbb{C}^* \rightarrow \mathbb{R}_{>0}$ be the map defined by $\phi(z) = |z|$. We have $\phi(zw) = |zw| = |z| \cdot |w| = \phi(z)\phi(w)$ and so ϕ is a homomorphism. If $r \in \mathbb{R}_{>0}$, then $r \in \mathbb{C}^*$ and $r = \phi(r)$, i.e. ϕ is surjective. Let K be the kernel of ϕ , then $K = \{z \in \mathbb{C}^* \text{ s.t. } \phi(z) = 1\} = \{z \in \mathbb{C}^* \text{ s.t. } |z| = 1\} = S^1$.

By the First Homomorphism Theorem, $\mathbb{C}^*/S^1 \cong \mathbb{R}_{>0}$.